CHAPTER 6

SYSTEM OF PARTICLES AND ROTATIONAL MOTION

CHAPTER ESSENCE AT A GLANCE:
Just like mass resists linear acceleration, Moment of Inertia (I) resists rotational spin! The farther the mass is spread from the spin axis, the harder it is to rotate. Rolling is simply linear translation plus rotational spinning working together!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Linear vs Rotational Analogues:** Mass m Moment of Inertia I; Force F Torque τ; Linear Momentum p Angular Momentum L; Velocity v Angular Speed ω.

• **Moment of Inertia (I = Σ m i r i^2):** Measures rotational inertia. Ring = M R^2; Disc = ½ M R^2; Solid Cylinder = ½ M R^2; Solid Sphere = M R^2. Radius of gyration k = sqrt(I divided by M).

• **Torque & Angular Momentum:** Torque τ = r × F = I α. Angular momentum L = r × p = I ω. Fundamental relation: τ = dL divided by dt (analogue of F = dp divided by dt).

• **Conservation of Angular Momentum:** If external torque τ ext = 0 implies L = I ω = constant implies $I_1 \omega_1 = I_2 \omega_2$. When a spinning figure skater folds arms, I decreases so angular speed ω shoots up!

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}, \quad \tau = \mathbf{r} \times \mathbf{F} = I\alpha$$

Center of Mass and Torque Vector Relation

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = I\omega, \quad \tau = \frac{d\mathbf{L}}{dt}$$

Angular Momentum and Newton's Rotational Analogue

$$K_{\text{rot}} = \tfrac{1}{2} I\omega^2, \quad I = \sum m_i r_i^2 = Mk^2$$

Rotational Kinetic Energy and Radius of Gyration

$$I_{\text{ring}} = MR^2, \quad I_{\text{disc}} = \tfrac{1}{2}MR^2, \quad I_{\text{sphere}} = \tfrac{2}{5}MR^2$$

Moments of Inertia for Standard Geometries

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A solid sphere rolls down an inclined plane FASTER than a disc or a ring because its moment of inertia ratio (k^2 divided by R^2 = 2 divided by 5 = 0.4) is the smallest!

• **Alert:** When a spinning skater pulls their arms inward, I decreases so angular speed ω instantly shoots up!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define centre of mass of a system of particles. Write the coordinates of the centre of mass of a two-particle system. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition]** Centre of mass is the unique point at which the entire mass of a system is assumed to be concentrated, moving as if all external forces acted directly at that single point. [1 Mark]

• **[Step 2: Two-Particle Coordinates]** For masses m1 and m2 at positions x1 and x2: X cm = (m1 × 1 + m2 × 2) divided by (m1 + m2). In vector form: R cm = (m1 r1 + m2 r2) divided by (m1 + m2). [1 Mark]

• **Key Exam Result:** R cm = (m1 r1 + m2 r2) divided by (m1 + m2)

2. Define Moment of Inertia (I) and Radius of Gyration (k) of a rigid body. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Moment of Inertia]** Moment of inertia measures the rotational inertia of a rigid body about an axis: $I = \Sigma m_i r_i^2 = \int r^2 dm$ (SI unit: kg m^2). [1 Mark]

• **[Step 2: Radius of Gyration]** Radius of gyration k is the distance from the rotational axis at which, if the total mass M were concentrated as a point mass, it would have the identical moment of inertia: $I = M k^2$ => k = sqrt(I divided by M) (SI unit: metre). [1 Mark]

• **Key Exam Result:** $I = \Sigma m r^2$, k = sqrt(I divided by M)

3. Derive the relation between torque and angular momentum: τ = dL divided by dt. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Angular Momentum Definition]** L = r × p. Differentiating with respect to time t: dL divided by dt = d divided by dt (r × p) = (dr divided by dt x p) + (r x dp divided by dt). [1 Mark]

• **[Step 2: Evaluating Cross Products]** Since dr divided by dt = v and p = m v: dr divided by dt x p = v x (m v) = m (v x v) = 0 (cross product of parallel vectors is zero). [1 Mark]

• **[Step 3: Final Relation]** Since dp divided by dt = F (Newton's second law): dL divided by dt = 0 + (r × F) = τ. Therefore: τ = dL divided by dt (Rotational analogue of F = dp divided by dt). [1 Mark]

• **Key Exam Result:** τ = dL divided by dt

4. Derive the relation between torque and angular acceleration: $\tau = I \alpha$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Angular Momentum and Moment of Inertia]** For a rigid body rotating with angular velocity ω about a fixed axis: $L = I \omega$. [1 Mark]

• **[Step 2: Time Differentiation]** Differentiating with respect to time t: dL divided by dt = d divided by dt (I ω) = I (d ω divided by dt). [1 Mark]

• **[Step 3: Torque Relation]** Since d ω divided by dt = α (angular acceleration) and dL divided by dt = τ: $\tau = I \alpha$. [1 Mark]

• **Key Exam Result:** $\tau = I \alpha$

5. Derive an expression for the rotational kinetic energy of a rigid body rotating with angular speed ω: K rot = ½ I ω^2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Particle Kinetic Energy]** Consider a rigid body composed of particles of masses m1, m2, ... at distances r1, r2, ... from the axis of rotation. Linear speed of ith particle is $v_i = r_i \omega$. Kinetic energy of ith particle: K_i = half (0.5) $m_i v_i^2$ = half (0.5) $m_i (r_i \omega)^2$ = half (0.5) $m_i r_i^2 \omega^2$. [1.5 Marks]

• **[Step 2: Summing Over All Particles]** Total rotational KE: $K_{rot} = \Sigma K_i$ = half (0.5) ($\Sigma m_i r_i^2$) ω^2. Since $I = \Sigma m_i r_i^2$: $K_{rot} = \frac{1}{2} I \omega^2$. [1.5 Marks]

• **Key Exam Result:** $K_{rot} = \frac{1}{2} I \omega^2$

6. A solid cylinder of mass 20 kg rotates about its axis with an angular speed of 100 rad/s. The radius of the cylinder is 0.25 m. Calculate the rotational kinetic energy and the angular momentum of the cylinder. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Moment of Inertia of Cylinder]** Cylinder about axis: I = $\frac{1}{2}$ M R^2 = half (0.5) × 20 x $(0.25)^2$ = 10 × 0.0625 = 0.625 kg m^2. [1 Mark]

• **[Step 2: Rotational Kinetic Energy]** K rot = $\frac{1}{2}$ I ω^2 = half (0.5) × 0.625 x $(100)^2$ = 0.3125 × 10,000 = 3125 J = 3.125 kJ. [1 Mark]

• **[Step 3: Angular Momentum]** L = I ω = 0.625 × 100 = 62.5 J s = 62.5 kg m^2 s^{-1}. [1 Mark]

• **Key Exam Result:** K rot = 3125 J, L = 62.5 kg m^2 s^{-1}

7. (a) What is radius of gyration? Give its SI unit. (1 Mark)n(b) Find the ratio of the radius of gyration of a circular ring to that of a circular disc of the same radius about their central transverse axes. (3 Marks) [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition & Unit]** Radius of gyration is the effective distance k from the rotation axis such that I = M k^2. SI unit: metre (m). [1 Mark]

• **[Step 2: Radius of Gyration of Ring]** For circular ring: I ring = M R^2 = M k $ring^2$ ⇒ k ring = R. [1.5 Marks]

• **[Step 3: Radius of Gyration of Disc & Ratio]** For circular disc: I disc = $\frac{1}{2}$ M R^2 = M k $disc^2$ ⇒ k disc = R divided by √2. Ratio k ring divided by k disc = R divided by [R divided by √2] = √2 divided by 1 = √2 : 1 ≈ 1.414 : 1. [1.5 Marks]

• **Key Exam Result:** k ring : k disc = sqrt(2) : 1